# Dual-Branch Melanoma Classification with Metadata Fusion, Focal Loss, and Mahalanobis OOD Detection

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***Abstract*—This work presents a dual-branch deep learning system fusing dermoscopic images with patient metadata (age, sex, anatomical site) for binary melanoma classification on the SIIM-ISIC 2020 dataset (13.6% malignant). Using an EfficientNet-B0 backbone with sigmoid focal loss ($\alpha$=0.864), a hyperparameter search across five experiments (Baseline, Experiments 1–4) isolates the effect of learning-rate scheduling, optimiser choice, and focal $\gamma$. Cosine annealing emerges as the most impactful change (F1: $0.8932 \rightarrow 0.8984$, false negatives reduced 12.5%). An image-only ablation (Experiment 4) confirms metadata fusion adds +1.9 percentage points (pp) recall; SHAP analysis identifies anatomical site and age as dominant predictors. With six-transform test-time augmentation (TTA) the system achieves F1 = 0.9134, recall = 91.48%, specificity = 98.62% (two-transform eval TTA on Baseline; Gradio demo uses six transforms). EigenCAM validates ABCDE-aligned attention and a Mahalanobis OOD detector flags non-dermoscopic inputs.**

***Index Terms*—melanoma, focal loss, metadata fusion, SHAP, test-time augmentation**

## I. Introduction

Melanoma causes over 75% of skin cancer deaths despite representing fewer than 5% of cases [1]. Five-year survival exceeds 98% at stage I but falls below 25% for metastatic disease [2]. Dermoscopy improves sensitivity by 10–27 pp [3], yet inter-observer variability persists. Esteva et al. [4] showed a fine-tuned CNN matched 21 dermatologists, and the ISIC 2020 challenge [6] released structured metadata alongside dermoscopic images, motivating late-fusion approaches that incorporate patient context. A further challenge is severe class imbalance: at ∼13.6% positive rate, a trivial all-benign predictor achieves 86.4% accuracy [11].

We select **EfficientNet-B0** [8] as the backbone because (i) the ISIC 2020 winning solution used an EfficientNet ensemble [7], (ii) compound scaling offers a strong accuracy-per-FLOP trade-off at only 5.3 M parameters, and (iii) its smooth convergence makes it well-suited for controlled hyperparameter (HP) studies (Section IV-C).

**Contributions:**

1) Dual-branch fusion (CNN + metadata MLP) with focal loss $\alpha$=0.864 derived from class proportions.
2) Five-experiment HP search (Baseline, Exp. 1–4) isolating scheduler, optimiser, focal $\gamma$, and metadata fusion, with per-experiment analysis of false-negative / false-positive trade-offs.
3) Measured ablation (+1.9 pp recall from metadata), per-feature SHAP analysis, error analysis, EigenCAM saliency, and Mahalanobis OOD.

**Out of scope:** Vision Transformers [17], [18] (require more data to compensate for lack of CNN inductive biases [17]), GAN augmentation, ensembles, multi-backbone comparison, multi-class diagnosis, and the ISIC private test set.

## II. Literature Review

### A. CNN Backbones for Dermoscopy

The ISIC 2020 challenge [6] released 33,126 dermoscopic images with patient metadata; our training set concatenates ISIC 2020 with additional ISIC 2019 images, yielding 37,648 samples. The winning solution used an EfficientNet ensemble [7] (AUC = 0.9490). **EfficientNet-B0** [8] applies compound scaling to balance accuracy and efficiency with only 5.3 M parameters. DenseNet-121 [9] and ResNet-50 [10] are popular alternatives but require more parameters for equivalent accuracy. ViTs [17], [18] show promise but need substantially more data due to lack of translation equivariance [17]; CNNs with ImageNet pretraining remain pragmatic at this scale.

### B. Class Imbalance and Focal Loss

With 6.4 benign examples per malignant, standard cross-entropy steers gradients toward the majority class so the network can ignore rare positives and still minimise loss. Focal loss [11] multiplies the per-example loss by $(1 - p_t)^\gamma$: confident, correct predictions contribute nearly zero gradient; uncertain or wrong predictions receive amplified signal, forcing attention onto hard malignant cases. Setting $\alpha$=1−0.136=0.864 further upweights malignant examples. Effect of varying $\gamma$ is in Section IV-B.

### C. Metadata Fusion, TTA, and Limitations

Pacheco et al. [12] curate a multimodal skin lesion dataset pairing clinical images with patient metadata, motivating fusion approaches. Groh et al. [13] caution about skin-tone bias. TTA [14] averages predictions over augmented views to reduce variance. Fujisawa et al. [15] identify amelanotic and small lesions as persistent DL failure modes. Davis and Goadrich [16] show AUC can overstate clinical utility on imbalanced data, motivating per-category error analysis (Section IV-E).

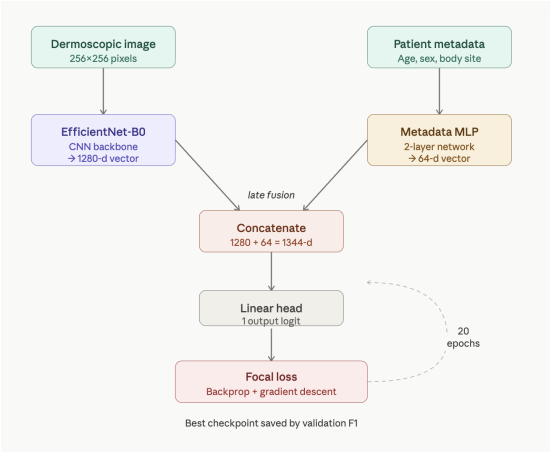


Fig. 1. Dual-branch pipeline: EfficientNet-B0 image branch and metadata MLP are fused via late concatenation, trained end-to-end with focal loss.

## III. Methodology

### A. Dataset

The SIIM-ISIC 2020 dataset contains ∼37 k dermoscopic images (256×256, **13.6% malignant**) with metadata: `age_approx` (continuous), `sex` (binary), and `anatom_site` (6 categories). An 80/20 stratified split (seed 42) preserves the class ratio. All experiments share this identical split.

### B. Preprocessing and Augmentation

**Images (train):** Albumentations (CLAHE, blur, 90° snaps) then torchvision v2 (flip, affine ±15°, colour jitter, random erasing, ImageNet normalisation). **Images (val):** resize + normalise only. **Metadata:** median imputation + $z$-score for age; mode imputation + one-hot for categoricals → **14-d** vector.

### C. Architecture

A dual-branch late-fusion model (Fig. 1): (1) an ImageNet-pretrained **EfficientNet-B0** backbone (→ 1280-d embedding) with the classification head removed; (2) a two-layer metadata MLP (14→128→64, BatchNorm + ReLU + Dropout 0.3). Late fusion is chosen because images and clinical scalars benefit from separate processing before being combined. The 1280-d and 64-d embeddings are concatenated (→ 1344-d) and projected to a single logit. Exp. 4 disables the metadata branch to isolate its contribution.

### D. Loss and Training

Sigmoid focal loss [11]:

$$FL(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

with $\alpha$=0.864, $\gamma$=2.0. We define five experiments (Table I), each introducing *one primary change* so every factor can be isolated: **Baseline** (Adam, fixed LR, $\gamma$=2.0): minimal-tricks reference. **Exp. 1** adds cosine annealing to settle into sharper minima (higher F1, no new FP). **Exp. 2** Adam→AdamW + weight decay $10^{-5}$ to curb minority-class overfitting. **Exp. 3** lowers $\gamma$ to 1.5, reducing suppression of easy negatives to recover specificity lost when Exp. 2 widened the decision boundary. **Exp. 4** disables metadata MLP (image-only ablation). The checkpoint with the highest validation F1 is retained.

TABLE I
HP CONFIGURATIONS AND BEST-CHECKPOINT RESULTS (EFFICIENTNET-B0). ALL RUNS: LR=$10^{-4}$, BATCH 32, 20 EPOCHS; WD=$10^{-5}$ FOR EXP. 2–3, ELSE 0. ΔFN: CHANGE IN FALSE NEGATIVES VS. PRIOR ROW; ‡VS. BASELINE.

| Config | Optim | Sched | $\gamma$ | Meta | F1 | Recall | FN (ΔFN) | FP |
|---|---|---|---|---|---|---|---|---|
| Baseline | Adam | None | 2.0 | ✓ | 0.8932 | 92.16% | 80 (–) | 145 |
| Exp. 1 (cosine) | Adam | Cosine | 2.0 | ✓ | **0.8984** | 93.14% | 70 (−10) | 145 |
| Exp. 2 (AdamW) | AdamW | Cosine | 2.0 | ✓ | 0.8855 | **93.54%** | 66 (−4) | 181 |
| Exp. 3 ($\gamma$1.5) | AdamW | Cosine | 1.5 | ✓ | 0.8968 | 92.75% | 74 (+8) | 144 |
| Exp. 4† (no meta) | Adam | None | 2.0 | — | 0.9000 | 90.30% | 99 (+19)‡ | 106 |

†Image-only ablation (no metadata MLP).

### E. Test-Time Augmentation

During evaluation, two geometric transforms (original and horizontal flip) are averaged to reduce orientation-specific variance. The Gradio demo extends this to six transforms (original, h-flip, v-flip, 90°/180°/270° rotation), exploiting the rotational symmetry of dermoscopic lesions. Evaluation-pipeline TTA impact is reported in Section IV-A.

### F. Evaluation

Five metrics are reported: **Recall** $= TP/(TP{+}FN)$, **Specificity** $= TN/(TN{+}FP)$, **PPV** $= TP/(TP{+}FP)$, **F1** (harmonic mean of PPV and recall), and **Accuracy** (reported but not optimised—an all-benign predictor achieves 86.4% yet catches zero cancers). Recall is primary: a missed melanoma costs far more than a false alarm.

## IV. Experiments

All experiments use PyTorch 2.x, identical data splits, augmentation pipelines, and evaluation code.

### A. Incremental Ablation

Exp. 4 (image-only, same HP as Baseline) isolates the metadata branch. Its accuracy (97.28%) and raw F1 (0.9000) look strong, but the confusion matrix reveals the cost: 99 missed melanomas versus Baseline's 80—19 more patients undetected. At 13.6% prevalence, 19 extra misses barely register across 7,530 examples, illustrating why accuracy overstates clinical utility on imbalanced data [16]. Metadata adds +1.9 pp recall (FN: 99→80) at the cost of 39 extra FP, as the MLP responds to anatomical-site cues absent from image features alone. Fig. 2 confirms Exp. 1 maintains a consistent recall lead over the image-only ablation across all 20 epochs.

Two-transform TTA (original + h-flip) applied to the Baseline model trades 7 extra FN for 55 fewer FP (recall 92.16%→91.48%, F1 = 0.9134, specificity 98.62%), tightening the decision boundary at a small sensitivity cost.

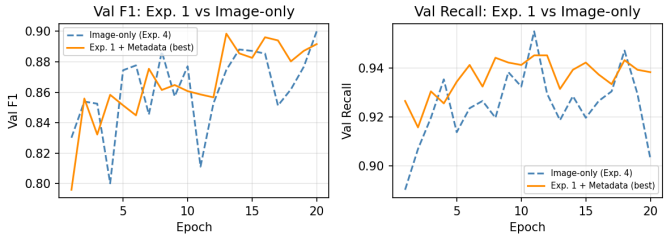


Fig. 2. Validation F1 and recall over 20 epochs: image-only (Exp. 4) vs. Exp. 1 (best model, with metadata MLP + cosine annealing). Metadata fusion provides a consistent recall advantage throughout training.

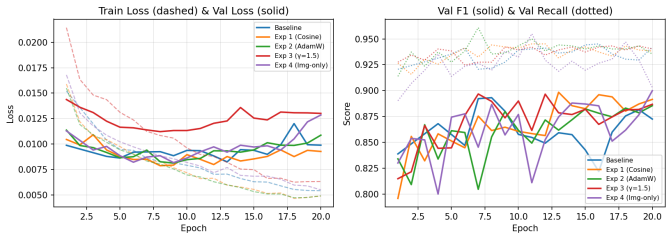


Fig. 3. Training and validation loss (left) and validation F1/recall (right) for all five experiments. Cosine annealing (Exp. 1–3) drives training loss lower; Exp. 4 (image-only) tracks Baseline but with lower F1, confirming the metadata contribution.

## B. Hyperparameter Search

The five experiments form a deliberate chain (Table I). **Exp. 1** (cosine annealing): FN drops 80→70 with *zero* extra FP ($\Delta$F1 = +0.0052) — the strongest single improvement. **Exp. 2** (AdamW): peaks recall at 93.54% but surges FP 145→181 ($\Delta$F1 = −0.0129), broadening the decision boundary. **Exp. 3** ($\gamma$=1.5): restores specificity — FP falls to 144 ($\Delta$F1 = +0.0113) while FN rises modestly to 74. Metadata fusion (Baseline vs. Exp. 4) adds +1.9 pp recall (19 fewer missed cancers) at the cost of 39 extra FP; the $\Delta$F1 of −0.0068 reflects increased FP, not worse cancer detection. Two-transform TTA on the Baseline adds $\Delta$F1 = +0.0202, trading 7 extra FN for 55 fewer FP.

## C. Training Dynamics

Fig. 3 overlays training dynamics. Baseline converges smoothly (loss 0.0152→0.0054, gap <0.32%). Cosine annealing (Exp. 1–3) drives loss lower but widens the train/val gap (0.36–0.59%); best-checkpoint selection by validation F1 mitigates this. Exp. 3 shows the largest gap; Exp. 1 achieves the best trade-off. All five experiments reach AUC = 0.99, confirming HP changes affect the precision–recall operating point rather than ranking ability [16].

## D. SHAP Feature Importance

SHAP KernelExplainer (Fig. 4) reveals the metadata feature importance for Exp. 1 (best model). `anatom_site_torso` is the dominant predictor; `age_approx` shows a monotonic relationship (older → higher risk); `anterior_torso` is rare but high-impact (SHAP up to 0.18). Sex contributes modestly. Feature rankings are consistent across all configurations;

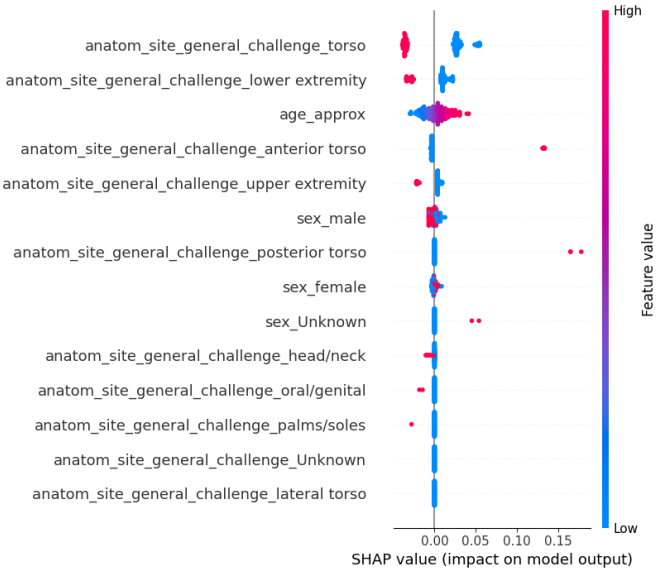


Fig. 4. SHAP beeswarm plot for Exp. 1 (best model). Feature importance is ranked by mean |SHAP value|; colour indicates feature magnitude (red = high, blue = low). `anatom_site_torso` and `age_approx` are the dominant predictors.

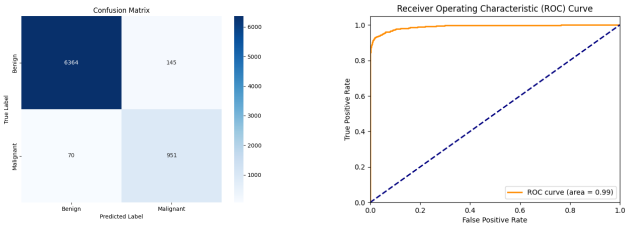


Fig. 5. Exp. 1 (best model): confusion matrix (left) and ROC curve (right, AUC = 0.99). The confusion matrix confirms 70 FN and 145 FP from 7,530 validation samples.

Exp. 3 ($\gamma$=1.5) is the only exception, compressing magnitudes and distributing importance more evenly.

## E. Error Analysis

**FN (∼7–8.5% of malignant):** amelanotic melanomas, small lesions (<6 mm), and early superficial spreading variants—subtypes lacking the pigmented features CNNs rely on [15]. **FP (∼1.4–2.8% of benign):** dysplastic naevi, atypical Spitz naevi, and irregular seborrhoeic keratoses [5]. Exp. 1 achieves a 6.86% FN rate (Fig. 5), acceptable for a screening tool.

## F. OOD Robustness

A Mahalanobis distance-based OOD detector [19] computes $\mu$ and $\Sigma^{-1}$ of the backbone's 1280-d validation features; inputs beyond $\mu+3\sigma$ are flagged. Thresholds range from 2637 (Exp. 3) to 3248 (Baseline); cosine-annealed models produce tighter feature distributions. Without this gate, non-dermoscopic inputs activate similar low-level CNN filters and align with malignant-class weights, yielding misleadingly high

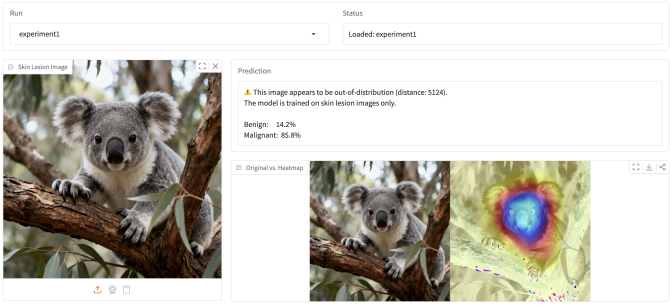


Fig. 6. OOD detection in the Gradio demo (Exp. 1). A koala photograph yields Mahalanobis distance 5124, far exceeding the threshold of 2652. The system issues an OOD warning; without it the unconstrained sigmoid head assigns 85.8% malignant confidence.

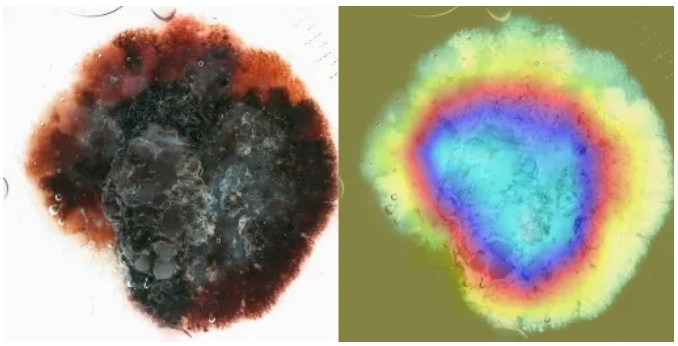

Fig. 7. EigenCAM overlay for a malignant lesion (Exp. 1). Left: original dermoscopic image. Right: activation heatmap—attention concentrates on asymmetric border and colour variegation, consistent with clinical ABCDE criteria [4].

confidence—Fig. 6 demonstrates a koala scoring distance 5124 ($\gg$ threshold 2652) yet receiving 85.8% malignant from the raw sigmoid. The detector intercepts at the embedding level before the logit is trusted.

### G. EigenCAM Interpretability

EigenCAM (Fig. 7) saliency maps confirm attention localises to asymmetric borders, colour variegation, and atypical pigment networks, aligning with clinical ABCDE criteria [4].

## V. Conclusion and Future Work

A dual-branch EfficientNet-B0 melanoma classifier fusing dermoscopic images with patient metadata achieves $F1 = 0.8984$, recall $= 93.14\%$ (Exp. 1, best epoch 13), reducing missed melanomas by 12.5% over Baseline with no additional FP. Two-transform TTA (original + h-flip) on the Baseline achieves $F1 = 0.9134$, recall $= 91.48\%$, specificity $= 98.62\%$ (trades 0.7 pp recall for 55 fewer FP; the Gradio demo extends this to six transforms). Ablation (Exp. 4) confirms metadata adds $+1.9$ pp recall; SHAP identifies `anatom_site_torso` and `age_approx` as dominant predictors across all HP experiments. EigenCAM validates ABCDE-aligned attention. A Mahalanobis OOD detector with empirically computed thresholds provides runtime safeguards against non-dermoscopic inputs.

**Deployment.** The full codebase is open-sourced at github.com/burrows99/melanoma-classification. Trained weights for all five experiments are hosted at huggingface.co/burrows99/melanoma-models, and a public Gradio demo is deployed at huggingface.co/spaces/burrows99/melanoma-classificiation, enabling browser-based inference with EigenCAM overlays and OOD detection.

**Future work:** (1) Larger EfficientNet variants (B3/B5) [7]; (2) multi-backbone comparison (DenseNet-121 [9], ResNet-50 [10]); (3) equity assessment via Fitzpatrick stratification [13]; (4) external validation on the ISIC private test set; (5) ViT/Swin backbones [17], [18] with larger pretraining budgets.

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